Reducing Alert Fatigue Through Risk-Based Cybersecurity Triage

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Final Project Presentation

Identifying the Problem

Problem Statement: Modern organizations generate thousands of cybersecurity alerts each day from security monitoring tools, but many of these alerts are false positives or low-risk events. Security analysts often lack effective data-driven methods to quickly prioritize alerts, making it difficult to distinguish real threats from noise. As a result, analysts spend significant time investigating non-critical alerts while high-risk incidents may be delayed or overlooked. This problem matters because inefficient alert triage increases the likelihood of data breaches, financial loss, operational disruption, and reputational damage. Security analysts and IT teams are directly affected through alert fatigue and resource strain, while the organization as a whole faces elevated cyber risk due to delayed or inaccurate threat detection.

Alert volume, false positives, analyst overload

Risk, Cost, Reputation

Initial Hypothesis

ASSUMED CORE ISSUE = ACCURACY OF ALERT LEGITIMACY

GOAL = CLASSIFY ALERTS AS REAL VS FALSE

FOCUSED ON FILTERING OUT NOISE

Root Cause Analysis– 5 Whys

Problem: *Analysts struggle to quickly identify real threats from false positives or low risk events, leading to alert fatigue and delayed responses to critical threats.*

Why is this happening?

Because security tools generate a high volume of alerts with limited prioritization or contextual scoring

Why is this happening?

Because many alerts are rule based and lack behavioral and risk-based enrichment

Why is this happening?

Because alert data is fragmented across multiple tools which makes it difficult to correlate signals into a single threat narrative.

Why is this happening?

Because analysts rely heavily on manual investigation and experience rather than automated data driven prioritization models.

Why is this happening?

Because organizations lack a scalable analytics framework that integrates asset criticality and threat intelligence to rank alerts by business risk.

Reframing the Problem

- PROBLEM IS NOT DETERMINING LEGITIMACY
- PROBLEM IS HOW THE ALERTS ARE TRIAGED
- ALL ALERTS SHOULD BE TREATED AS POTENTIALLY LEGITIMATE
- PRIORITY SHOULD BE RISK-BASED, NOT BINARY

Six Thinking Hats– *Evaluating* Alert Triage

(1)White - What are the facts?	(2)Green - What are some potential solutions?	(3)Yellow - What are the upsides of the proposed solutions?	(4)Black - What are the downsides of the proposed solutions?	(5) Red - What does our intuition tell us?	(6) Blue - What's our decision?
Alert data, volumes, tools	Automation, risk scoring	Efficiency, reduced fatigue	False confidence, model bias	Analyst Frustration	Risk–based triage

Challenging Assumptions with What-If & Start-Stop-Continue

What-If Analysis

What: Assume all alerts are valid; focus on faster, smarter triage

Why: Deciding "real vs false" slows response and increases alert fatigue

What If:

- Best case: High-risk alerts prioritized immediately
- Worst case: Poor triage still buries critical incidents
- Likely case: Analysts focus on impact, not noise

How:

- Prioritize by risk, asset value, and context
- Automate scoring to accelerate response

Start-Stop-Continue

Start:

- Prioritizing alerts assuming all are legitimate
- Using risk, context, and impact to triage alerts faster and more accurately
- Optimizing analyst workflows around speed and severity, not validation

Stop:

- Treating "is this a real alert?" As the primary decision point
- Spending analyst time proving alerts are false positives
- Delaying response while debating alert legitimacy

Continue:

- Leveraging data-driven scoring to rank alerts by urgency
- Focusing analyst effort on the highest-risk alerts first
- Reducing alert fatigue through smarter, faster triage

SWOT Analysis– Risk–Based Triage

Strengths	Scalable & Efficient
Weaknesses	Model dependency
Opportunities	AI + Automation
Threats	Evolving threats & Data quality

AI Agent – Supporting Risk-Based Triage

- Multi-step AI-assisted workflow
- Mirrors analyst thinking
- Enforces structured analysis
- Requires human approval at key points
- Supports risk-based triage (not alert dismissal)

Impact, Action & Conclusion

REDUCED ALERT FATIGUE

FASTER RESPONSE TO REAL THREATS

BETTER ALIGNMENT WITH BUSINESS RISK

CLEAR NEXT STEPS FOR IMPLEMENTATION